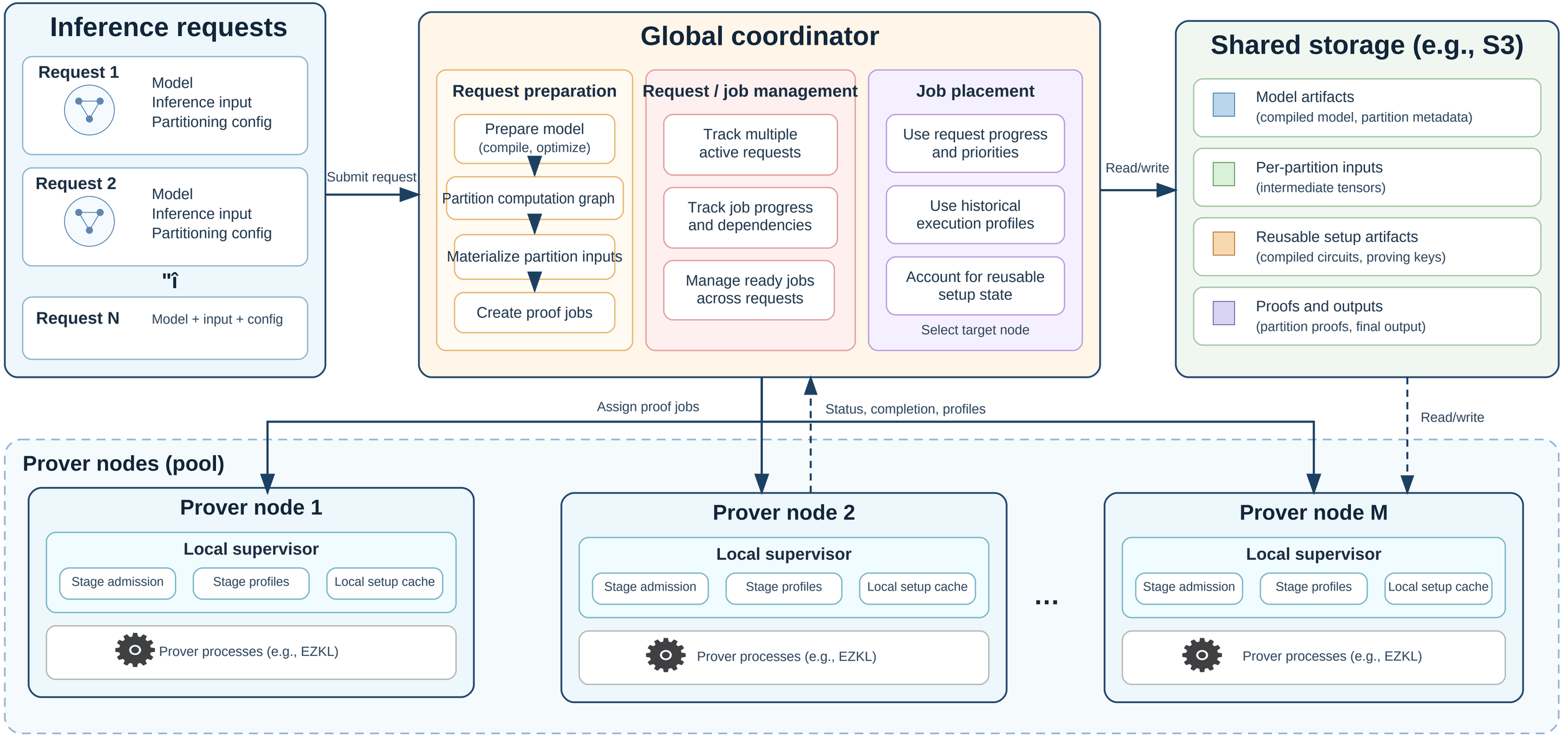
Inference requests
Request 1
Model
Inference input
Partitioning config
Request 2
Model
Inference input
Partitioning config
"î
Request N
Model + input + config
Submit request
Global coordinator
Request preparation
Prepare model
(compile, optimize)
Partition computation graph
Materialize partition inputs
Create proof jobs
Request / job management
Track multiple
active requests
Track job progress
and dependencies
Manage ready jobs
across requests
Job placement
Use request progress
and priorities
Use historical
execution profiles
Account for reusable
setup state
Select target node
Read/write
Shared storage (e.g., S3)
Model artifacts
(compiled model, partition metadata)
Per-partition inputs
(intermediate tensors)
Reusable setup artifacts
(compiled circuits, proving keys)
Proofs and outputs
(partition proofs, final output)
Assign proof jobs
Status, completion, profiles
Read/write
Prover nodes (pool)
Prover node 1
Local supervisor
Stage admission
Stage profiles
Local setup cache
Prover processes (e.g., EZKL)
Prover node 2
Local supervisor
Stage admission
Stage profiles
Local setup cache
Prover processes (e.g., EZKL)
...
Prover node M
Local supervisor
Stage admission
Stage profiles
Local setup cache
Prover processes (e.g., EZKL)